SOFTWARE PRODUCT DESIGN / V1.0

ENTERPRISE VAT ANGULAR FRONTEND

Professional Screen Design & Implementation Blueprint

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12 BUSINESS MODULES

DYNAMIC MENU + RBAC

MAKER-CHECKER-APPROVER

MICROSERVICES INTEGRATION

Prepared for Bangladesh VAT / Manufacturing / Trading / Service

01 / Solution Overview

Frontend principles and integration topology

Angular SPA

Standalone components, lazy routes, signals, reactive forms, bilingual UI.

API Gateway

Single public endpoint, OIDC, correlation ID, centralized routing.

ASP.NET Core Services

Identity, purchase, sales, stock, manufacturing, VAT, compliance, reporting.

Frontend Design Rules

Feature-first folders and shared design system.

Server-side authorization; hiding menus is not security.

Company/branch/period context on all business screens.

Recommended Stack

Angular + PrimeNG + SCSS; Angular Signals/RxJS.

OIDC Authorization Code + PKCE, no localStorage access tokens.

Playwright E2E + unit tests + accessibility checks.

02 / Complete Screen Inventory

12 modules / 80+ proposed screens; prioritize MVP first

01 Dashboard

Executive, VAT KPIs, Pending Work, Alerts

02 Administration

Company, BIN, Branch, User, Role, Menu, Matrix

03 Master Data

Customer, Supplier, Product, HS, UOM, Tax Code

04 Purchases

PO, GRN, Local Purchase, Import, Return, 6.1

05 Sales

Order, Invoice, 6.3, Return, Credit/Debit

06 Manufacturing

BOM, Coefficient 4.3, Production, Wastage

07 Inventory

Receipts, Issues, Transfers 6.5, Ledger

08 VAT & Compliance

Tax Ledger, VDS 6.6, Adjustments, 9.1

09 Approvals

Inbox, Verify, Approve, Reject, History

10 Reports

6.1, 6.2, 6.3, 9.1, Audit, MIS

11 Integrations

ERP Connections, Import, Sync Errors, Webhooks

12 Settings

Localization, Templates, Fiscal Lock, Preferences

03 / Screen Mockup: Executive Dashboard

Responsive shell, KPIs, month comparison, work queue

VAT ENTERPRISE

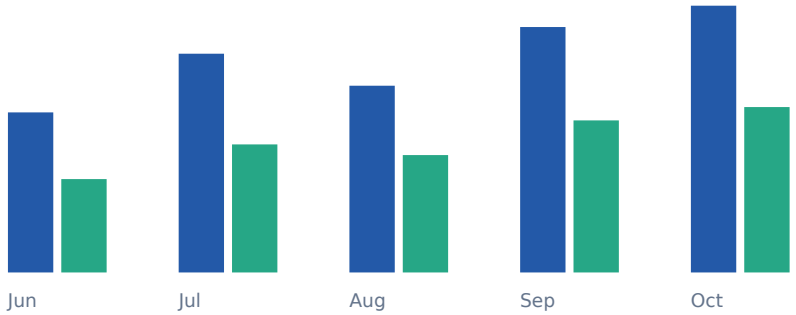
- Dashboard
- Administration
- Master Data
- Purchases
- Sales & Billing
- Manufacturing
- Inventory
- VAT & Compliance
- Approvals
- Reports
- Integrations

Home / Dashboard

Admin | EN

Total Sales	Total Purchases	Output VAT	Input VAT
BDT 12.0M	**BDT 8.5M**	**BDT 1.8M**	**BDT 1.05M**

Input vs Output VAT



Action Required

12 invoices to verify

4 returns to approve

2 sync exceptions

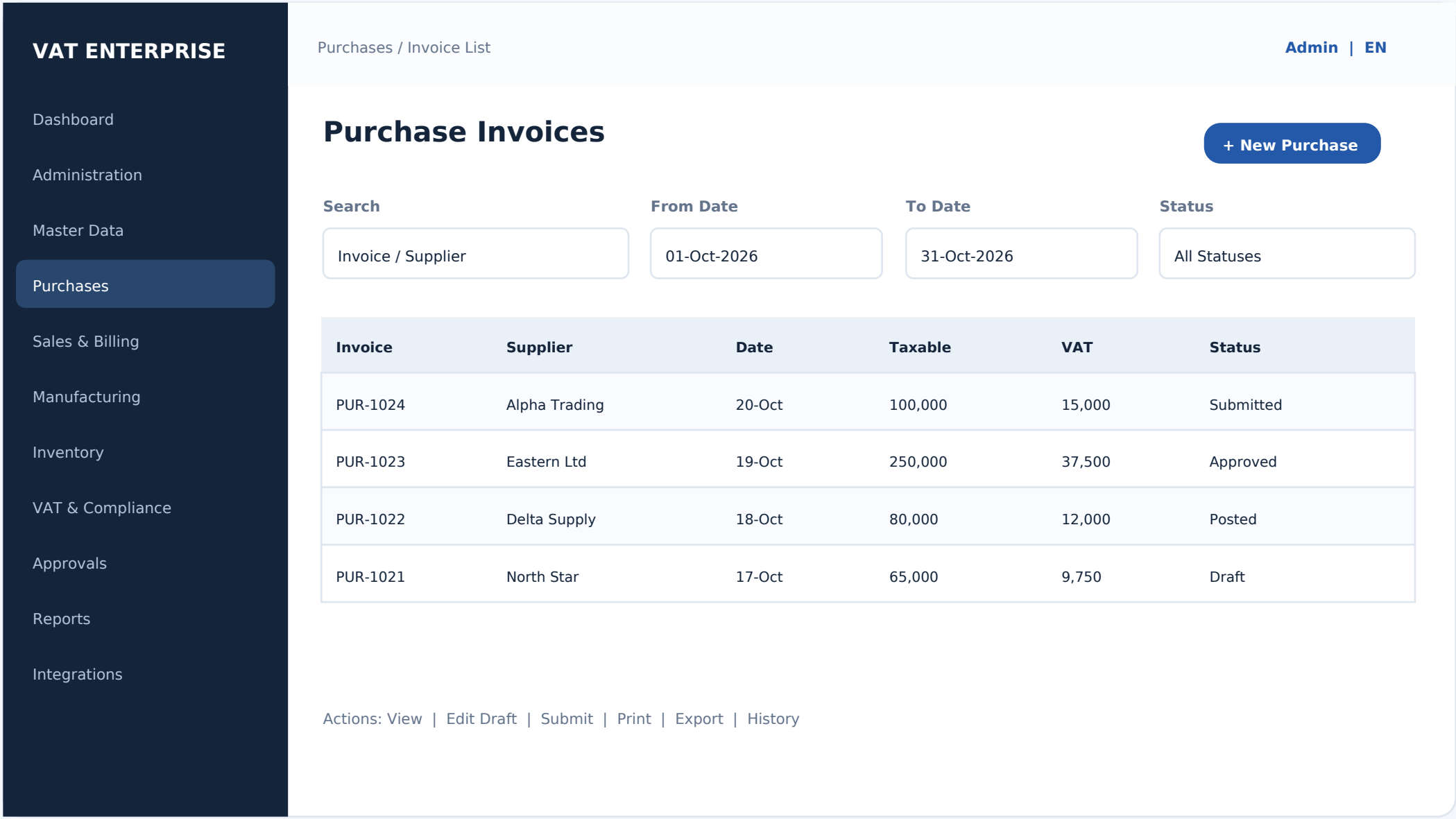
Period closing: Oct

Filters: Company | Branch | Fiscal Period | Tax Type

All figures are illustrative placeholders.

04 / Screen Mockup: Purchase Invoice List

Search, filters, approval status, export, row actions



05 / Screen Mockup: Purchase Invoice Entry

Header + line items + VAT preview + document actions

VAT ENTERPRISE

- Dashboard
- Administration
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- Integrations

Purchases / New Invoice

Admin | EN

Create Purchase Invoice

Save Draft

Supplier: Alpha Trading

Invoice Number: SUP-INV-889

Invoice Date: 23-Oct-2026

Branch: Dhaka Factory

Warehouse: Main Store

Tax Treatment: Standard / Configured

Product	Qty	Unit Price	Taxable	VAT	Total
Raw Material A	100	1,000	100,000	15,000	115,000
Packaging B	50	200	10,000	1,500	11,500

Taxable: 110,000 **VAT: 16,500** **Grand Total: 126,500**

Attachments | Validate VAT | Submit for Approval

06 / Screen Mockup: Sales Invoice / Mushak 6.3

Sales flow and printable tax invoice

VAT ENTERPRISE

- Dashboard
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Sales / Invoice Entry

Admin | EN

Sales Invoice / VAT Challan

Preview 6.3

Customer: Retail Distributor

Sales Order: SO-2026-155

Challan Date: 23-Oct-2026

Delivery Branch: Dhaka

Warehouse: Finished Goods

Supply Type: Local Standard

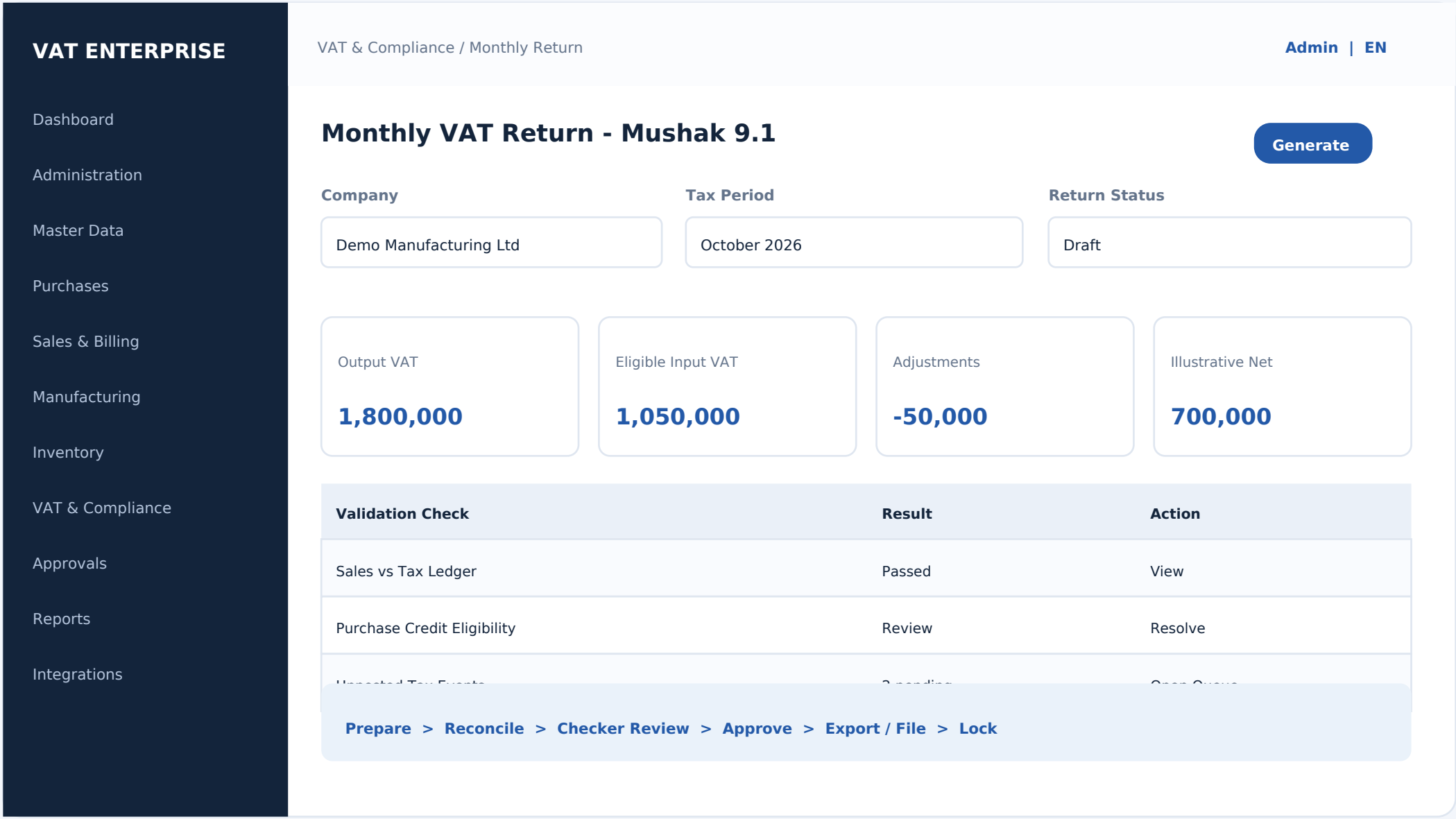
Item	Quantity	Unit Price	Base Value	VAT	Gross
Finished Product A	40	3,000	120,000	18,000	138,000
Service Charge	1	5,000	5,000	750	5,750

Base: 125,000 **Output VAT: 18,750** **Total: 143,750**

Actions: Save Draft | Submit | Approve | Post | Print | Credit Note

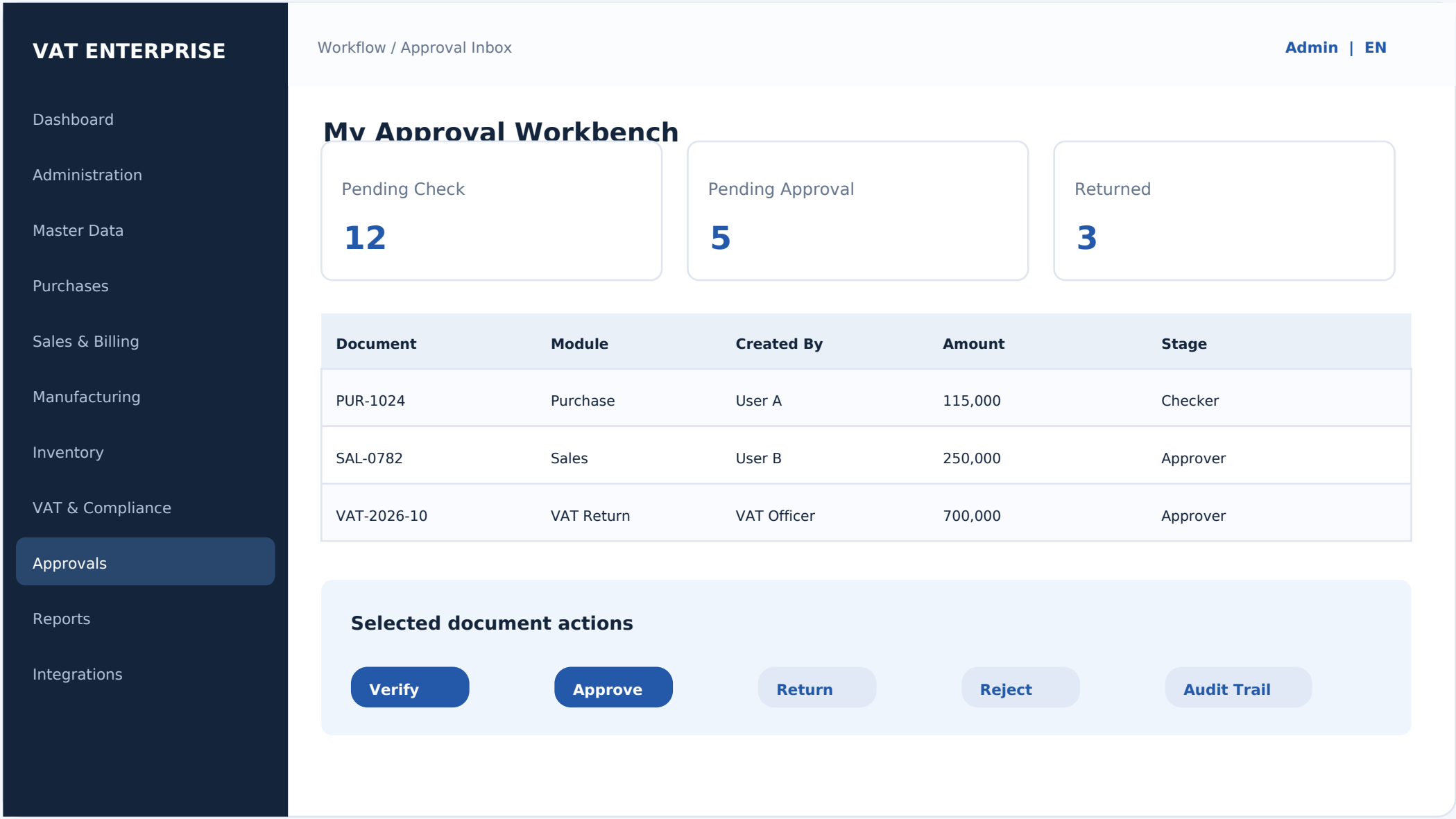
07 / Screen Mockup: Monthly VAT Return

Mushak 9.1 draft, reconciliation and approval controls



08 / Screen Mockup: Approval Workbench

Maker / checker / approver task inbox



09 / Roles, Permissions & Dynamic Menu

Configuration-driven navigation and server-side authorization

Menu Entity

id, parentId, code, labelKey, route, icon, sortOrder, featureFlag, active.

GET /api/v1/navigation/my-menu returns a filtered menu tree.

Permission Entity

permissionCode e.g. purchase.invoice.create / approve.

Role -> permissions; user -> roles; branch/company scope.

Role	Purchase	Sales	VAT Return	Approvals	Admin
System Admin	Config	Config	View	Config	Full
Purchase Maker	Create	-	-	Submit	-
Checker	Verify	Verify	Review	Check	-
VAT Manager	View	View	Prepare	Approve	-
Auditor	Read	Read	Read	Read	-

Rules: least privilege | maker != approver | branch scoping | server validates every action

10 / Angular Route Architecture

Lazy feature routes with canMatch / canActivate permission checks

URL Path	Lazy Page Component	Required Permission
/dashboard	DashboardPage	dashboard.view
/admin/users	UserListPage	identity.user.read
/admin/menus	MenuBuilderPage	identity.menu.manage
/master/products	ProductListPage	master.product.read
/purchases/invoices	PurchaseInvoiceListPage	purchase.invoice.read
/purchases/invoices/new	PurchaseInvoiceFormPage	purchase.invoice.create
/sales/invoices	SalesInvoiceListPage	sales.invoice.read
/inventory/stock-ledger	StockLedgerPage	inventory.ledger.read
/manufacturing/bom	BomListPage	manufacturing.bom.read
/vat/returns	VatReturnListPage	compliance.return.read
/workflow/inbox	ApprovalInboxPage	workflow.task.read
/reports/mushak-6-1	PurchaseRegisterReportPage	reports.mushak61.read

11 / Frontend Project Folder Structure

Maintainable feature-based architecture

```
src/app/
  core/auth/
  core/guards/
  core/interceptors/
  core/api/
  core/context/
  layout/shell/
  shared/ui/
  shared/forms/
  shared/pipes/
  features/dashboard/
  features/administration/
  features/master-data/
  features/purchases/
  features/sales/
  features/manufacturing/
  features/inventory/
  features/vat-compliance/
  features/workflow/
  features/reports/
  features/integrations/
```

Reusable Components

DataTable, SearchFilter, DateRange, MoneyInput, TaxRateSelector, DocumentStatus, AttachmentUploader, ApprovalTimeline, ConfirmDialog.

Per Feature Structure

pages/ | components/ | data-access/ | models/ | routes.ts | facade.ts | validators.ts | tests/

Do not put business-specific code in shared/.

12 / Approval & Document Lifecycle

A common workflow applied across transactional screens

01 Draft

Maker creates and edits

02 Submitted

Maker submits for verification

03 Checked

Checker validates details

04 Approved

Approver authorizes

05 Posting

Async services process transaction

06 Posted

Ledger and inventory confirmed

Exception paths: Returned -> Draft correction; Rejected -> Closed; PostingFailed -> Retry / reconcile.

Only server-confirmed posting success can display the final Posted status.

13 / Gateway Integration & Frontend Contracts

Microservices-aware Angular data access

Gateway Endpoints

/api/v1/auth/* and /navigation/*

/api/v1/purchases/* and /sales/*

/api/v1/inventory/* and /manufacturing/*

VAT & Compliance

/api/v1/tax/* and /compliance/*

/api/v1/workflows/* and /reports/*

/api/v1/integrations/*

Angular Http Interceptors

Access token or BFF session handling

X-Correlation-Id; standardized error parsing

401 -> login / renew; 403 -> access denied

Timeout and safe retry only for idempotent calls

Frontend Reliability

Pagination, filter and sort on server

Optimistic UI only for nonfinancial actions

Poll or subscribe to posting status

Persist drafts through backend, not browser only

Show request ID on supportable errors

14 / Delivery Plan & Acceptance Checklist

Recommended implementation sequence and measurable quality gates

Sprint 1-2	Design system, shell, OIDC, dynamic menu
Sprint 3-4	Admin, company context, master screens
Sprint 5-7	Purchase, import, sales, tax invoice
Sprint 8-9	Inventory, manufacturing and BOM
Sprint 10-12	VAT ledger, returns, approval workbench
Sprint 13-14	Reports, integrations, E2E and hardening

Acceptance: RBAC enforced | responsive | keyboard accessible | accurate tax display | audit trail | UAT signed off

15 / Implementation Notes & Compliance Boundaries

Engineering decisions before production rollout

Regulatory Accuracy

Mushak form fields, rate schedules, exemptions and filing rules must be validated against current NBR rules.

PDF mocks are proposed UI screens, not screenshots of a deployed product.

Performance & Security

Lazy routes, virtual scroll, server-side tables.

OIDC + PKCE, strict CSP, CSRF strategy as applicable.

Never trust client role, company or branch claims without server validation.

Definition of Done

API contract approved; validation complete.

Loading, empty, error and unauthorized states.

Responsive screen; tests; accessibility; audit events.

Recommended Next Deliverables

Figma-grade screen specification for every form.

OpenAPI contracts and permission catalog.

Angular starter project with working routes and UI.

PostgreSQL + backend integration tests.